TEST CODE **01207020**

FORM TP 2008002

JANUARY 2008

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

BIOLOGY

Paper 02 – General Proficiency

$1\frac{1}{2}$ hours

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. **Answer ALL questions on this paper.**
2. **Use this answer booklet when responding to the questions. For EACH question, write your answer in the space provided and return the answer booklet at the end of the examination.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

Copyright © 2006 Caribbean Examinations Council®.
All rights reserved.

Answer ALL questions.

DO NOT spend more than 30 minutes on Question 1.

1. A student plotted the graph shown in Figure 1 after carrying out an investigation on the effect of temperature on the rate of enzyme reaction.

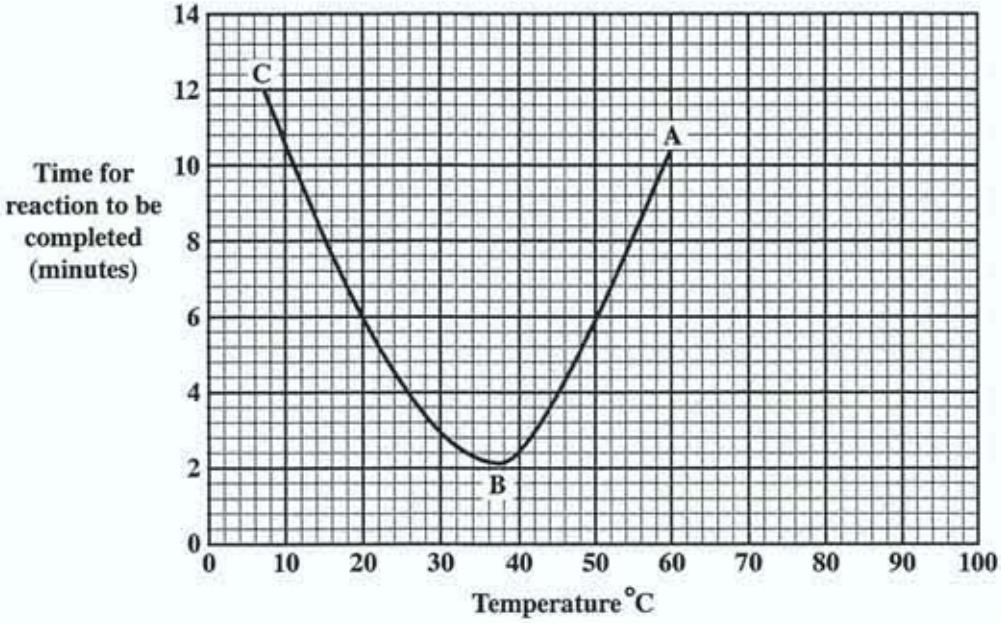


Figure 1. Graph showing the effect of temperature on the rate of enzyme activity

(a) (i) Describe the reaction rate between points A and B, and between points B and C in Figure 1.

Between A and B

_____

_____

Between B and C

_____

_____

(2 marks)

(ii) What is the optimum temperature for the enzyme in this reaction?

_____

(1 mark)

GO ON TO THE NEXT PAGE

(b) Using information in Figure 1, construct in the space provided below, a table to show the data obtained by the student in investigating the rate of enzyme reaction. Your table should contain six rows of data/information.

(8 marks)

(c) (i) If the enzyme used in this investigation were amylase, describe fully the method you would use to obtain the data for the graph. Identify the materials and apparatus.

Materials and apparatus

_____

_____

Method

_____

_____

_____

_____

(ii) Identify ONE source of error in the investigation.

_____

(7 marks)

GO ON TO THE NEXT PAGE

(d) Green papaya (paw paw) fruits may be used as a means of softening meat. The green papaya is crushed and rubbed on the meat and then left for a few hours before cooking.

(i) What process might be taking place to soften the meat?

_____

(1 mark)

If the crushed papaya is added to the meat while cooking has started, the cooking time for the meat is not significantly shorter.

(ii) Suggest why this may be so.

_____

_____

(2 marks)

(e) In order to germinate, seeds also require a softening agent. In this case they are softened by water.

(i) Suggest TWO ways in which water helps the process of germination.

_____

_____

(2 marks)

(ii) Apart from water, state TWO environmental conditions needed for germination.

_____

_____

(2 marks)

(f) Environmental conditions impact not only on germination of seeds but also on the functioning of soil inhabitants, including microorganisms.

(i) Name TWO important groups of microorganisms whose activities are influenced by environmental conditions.

_____

(2 marks)

(ii) Identify ONE cycle which depends on the activities of microorganisms.

_____

(1 mark)

(iii) Explain the importance of the microorganisms named in (f)(i) in natural cycles.

_____

_____

(2 marks)

Total 30 marks

GO ON TO THE NEXT PAGE

2. In conducting an ecology study, a group of Biology students investigates gaseous exchange in pond snails and in a waterweed species. The students set up five test tubes as shown in Figure 2. All tubes are filled with an indicator solution, stoppered with rubber bungs and exposed to light for one hour.

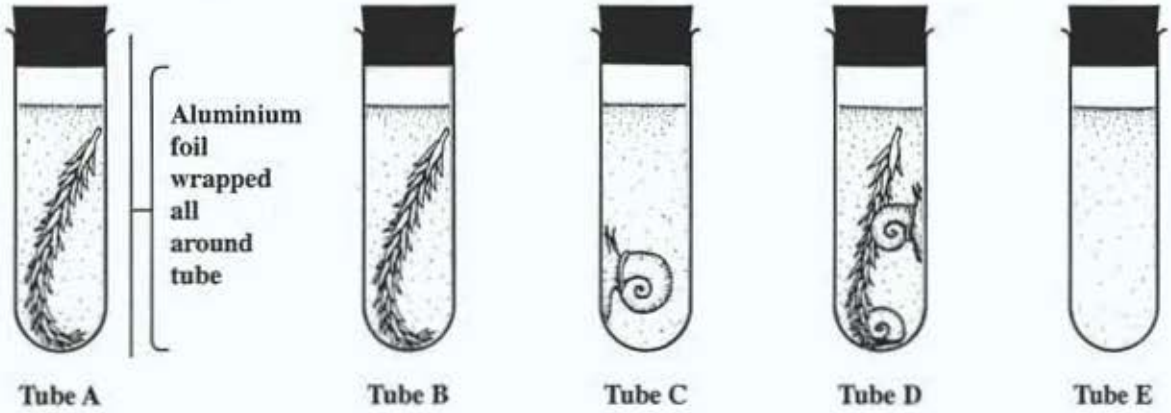


Figure 2. Set up of test tubes for investigating gaseous exchange in two species

The students use the colour changes in hydrogen carbonate indicator solution as shown in Table 1 to determine the carbon dioxide concentration in each tube.

TABLE 1

CARBON DIOXIDE CONCENTRATION AND BICARBONATE INDICATOR

Carbon dioxide concentration	Colour of bicarbonate indicator solution
Same as in the air	Reddish brown
Less than that in the air	Purple
More than that in the air	Yellow

(a) Identify ONE tube in which, after one hour, the colour of the hydrogen carbonate indicator becomes **yellow** and ONE in which the solution becomes **purple**. In each case explain your answer.

Tube with **yellow** indicator:

_____

Explanation:

_____

_____

3. Figure 3 shows a longitudinal section through the human heart.

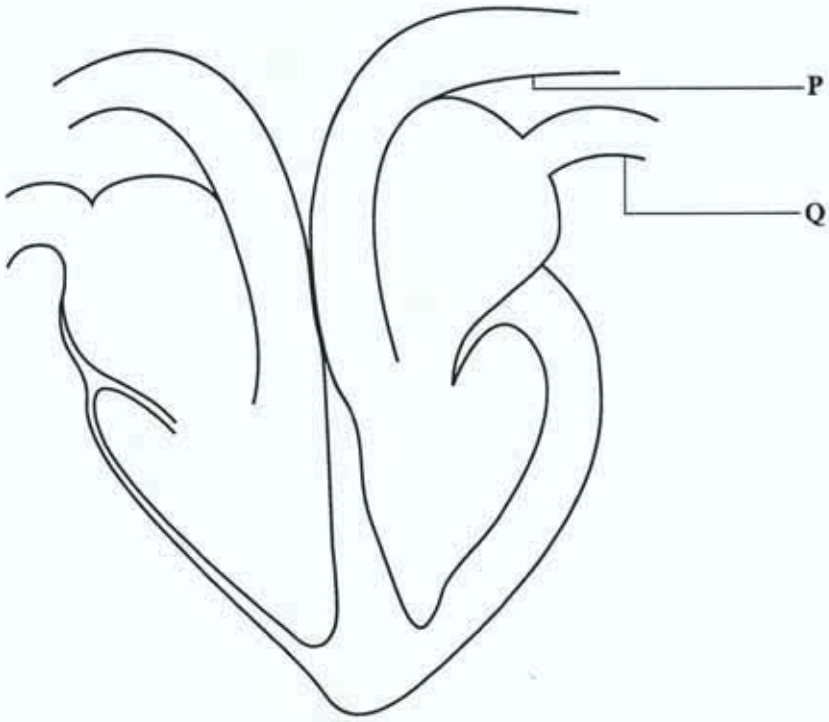


Figure 3. Diagram of a longitudinal section through the human heart

(a) (i) Using FOUR arrows ONLY show the direction of flow of blood into and out of the human heart illustrated in Figure 3. **(4 marks)**

(ii) Name the blood vessels labelled P and Q respectively.

Blood vessel P:_____

Blood vessel Q:_____

(2 marks)

(iii) Some babies are born with a small opening in the wall/septum between the two upper chambers of the heart. Suggest how this condition would affect the functioning of the heart when the atria contract.

_____

_____

(2 marks)

(b) Heart muscle cells obtain their requirements from blood transported in the coronary arteries.

(i) Name TWO substances that the heart muscle cells will receive in this blood supply.

_____

_____

(2 marks)

(ii) Identify ONE substance found in the blood supply that can cause the walls of the coronary artery to thicken.

_____

(1 mark)

(iii) How does the thickening of the coronary arterial wall affect the heart?

_____

_____

(2 marks)

(c) Plant cells obtain their requirements from transport vessels with which they are associated. Identify TWO types of transport vessels in plants and for EACH identify ONE substance which it transports.

Transport vessel 1:_____

Substance transported:_____

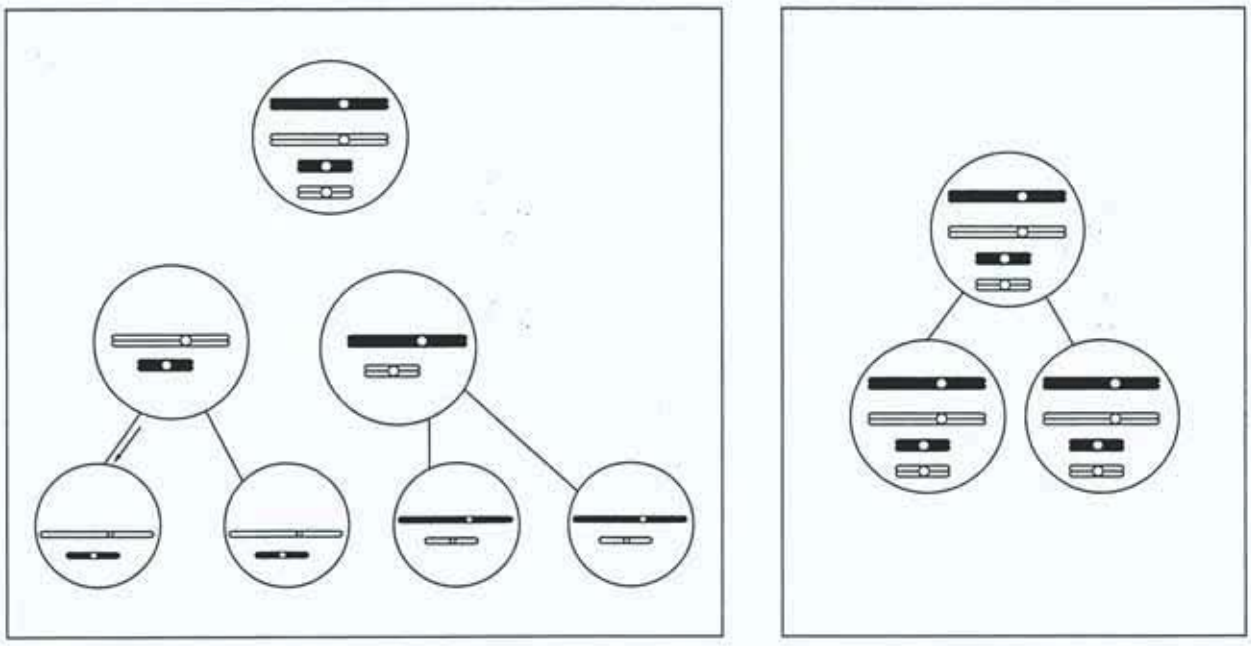
Transport vessel 2:_____

Substance transported:_____

(4 marks)

Total 17 marks

4. Figure 4 below shows two cells undergoing cell division.



Cell division I **Cell division II**

Figure 4. Cell division in two cells

(a) (i) Name EACH type of cell division illustrated in Figure 4.

Cell division I: _____

Cell division II: _____

(2 marks)

(ii) Identify TWO differences between the processes observed in the illustrations in Figure 4.

_____

_____

(2 marks)

(iii) Name TWO specific locations in a plant where EACH type of cell division illustrated in Figure 4 occurs.

Cell division I: _____

Cell division II: _____

(2 marks)

(iv) Give TWO reasons why BOTH types of cell division are important in the life cycle of a plant.

Cell division I:

_____

_____

Cell division II:

_____

_____

(4 marks)

(b) After egg cells from a female rabbit had undergone one form of cell division their nuclei were replaced with nuclei from somatic (body cells) of the rabbit. These cells produced offspring that were identical to the female rabbit.

(i) What is the name given to this process?

_____

(1 mark)

(ii) Suggest ONE advantage and ONE disadvantage of this process.

Advantage:

_____

Disadvantage:

_____

(2 marks)

(c) Egg cells are found in both plants and animals. Egg cells in plants are protected within an ovule.

(i) Into what structure will the ovule develop?

_____

(1 mark)

(ii) Name TWO features which will show that the structure you identified at (c)(i) developed from an ovule.

_____

_____

(2 marks)

Total 16 marks

GO ON TO THE NEXT PAGE

5. Figure 5 below shows the nitrogen cycle.

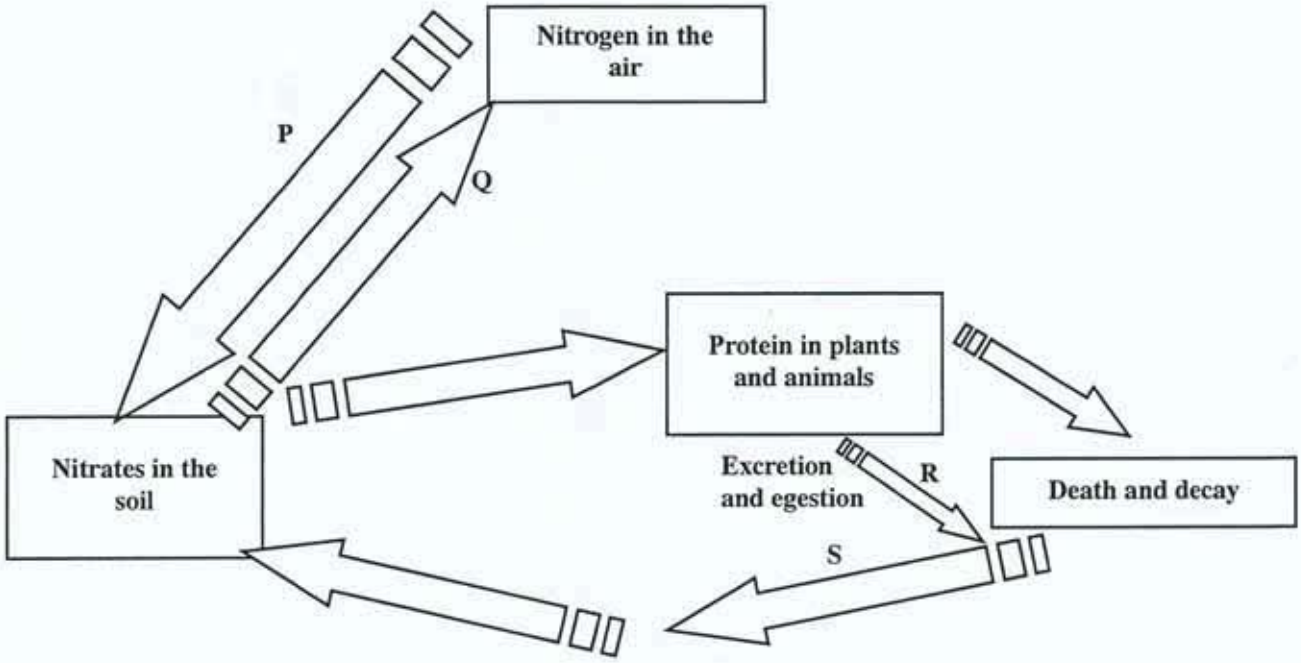


Figure 5. An incompletely labelled nitrogen cycle

(a) (i) Name the process carried out by bacteria at EACH of the stages indicated by P, Q, R and S.

P: _____

Q: _____

R: _____

S: _____

(4 marks)

(ii) Explain how nitrates from the soil become protein in plants and animals.

_____

_____

_____

(3 marks)

GO ON TO THE NEXT PAGE

(iii) Leguminous plants have bacteria in their root nodules, which are involved in the nitrogen cycle. The relationship between the legumes and the bacteria is described as mutualistic. Explain why this description is appropriate.

_____

_____

_____

(2 marks)

(b) Apart from mutualism, identify TWO OTHER types of relationships that occur among living organisms. Give ONE example of each relationship you identify.

Relationship: _____

Example: _____

Relationship: _____

Example: _____

(4 marks)

Total 13 marks

END OF TEST